
Replication and Predictable-Drift Diagnostics for FTS-Diffusion

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Abstract

1 Financial time-series generators are often evaluated by marginal stylized facts and
2 downstream forecasting gains, but those criteria can miss protocol artifacts and
3 artificial conditional mean predictability. We study FTS-Diffusion, an ICLR 2024
4 model based on scale-invariant subsequence clustering, pattern-conditioned diffu-
5 sion, and learned pattern evolution. Our replication finds that S&P 500 Train-on-
6 Augmentation-Test-on-Real (TATR) conclusions depend strongly on the synthetic
7 rollout: a six-seed continuous-rollout protocol reaches a mean best improvement
8 of 84.9%, whereas independent 252-day blocks never improve over baseline and
9 finish 193.2% worse. We also fail to reproduce the Appendix B.3 SISC toy targets,
10 with best average per-unit DTW 0.0321 versus 0.01 and best interval IoU 0.6418
11 versus 0.784. As an extension, we implement a fixed predictable-drift diagnos-
12 tic. On real S&P 500, GOOG, and ZC=F holdouts, the statistic lies inside asset-
13 specific block-shuffle null bands. In a 500-seed hidden-drift experiment, the same
14 statistic tracks exploitable artificial alpha: at sign persistence $p = 0.85$, mean out-
15 of-sample Sharpe is 7.72 versus 0.09 for an iid-sign control. When incorporated as
16 a 1000-seed augmentation-selection penalty, it lowers held-out normalized MSE
17 from 1.0452 to 1.0087 versus marginal stylized selection and nearly eliminates
18 high-persistence choices. The main lesson is that realistic-looking financial paths
19 are not enough: synthetic finance benchmarks need protocol-explicit generation
20 and direct checks for artificial alpha.

21 1 Introduction

22 Deep generative models are attractive for finance because historical market data are scarce, noisy,
23 nonstationary, and not experimentally reproducible. Financial returns also exhibit heavy tails, weak
24 raw-return autocorrelation, volatility clustering, and recurring shapes at irregular time scales [Cont,
25 2001]. FTS-Diffusion [Huang et al., 2024] addresses the last property directly by discovering
26 variable-length motifs and generating new trajectories motif by motif.

27 This paper is a replication and diagnostic-extension study. We do not introduce a replacement gener-
28 ator. Instead, we test whether the published downstream gains are robust to rollout details, whether
29 the SISC toy-data targets are reproducible, and whether a simple predictable-drift statistic catches
30 artificial alpha that marginal stylized facts may miss. The answer is mixed: the architecture is well
31 motivated, but the downstream evidence is fragile because continuous Markov-chain rollouts and
32 independent yearly blocks answer different questions.

33 We therefore extend the evaluation with a predictable-drift diagnostic. Under a fixed information
34 vector

$$\omega_t = (1, r_t, r_{t-1}, |r_t|, |r_{t-1}|)^T, \quad (1)$$

we measure whether the next return r_{t+1} has a nonzero linear conditional mean under a bounded test class. This is not a proof of market efficiency or absence of arbitrage. It is a finite, reproducible check for an important failure mode: a synthetic generator can match tails and volatility clustering while introducing a simple, exploitable conditional drift.

Contributions. We make four contributions: a replication summary on S&P 500, GOOG, and corn futures (ZC=F); a six-seed isolation of TATR rollout sensitivity; a documented Appendix B.3 SISC replication gap; and a predictable-drift diagnostic calibrated on real assets, validated by a 500-seed predictive-alpha experiment, and incorporated in a 1000-seed augmentation-selection benchmark.

2 Background: FTS-Diffusion

FTS-Diffusion decomposes financial time-series generation into three modules [Huang et al., 2024]. Scale-invariant subsequence clustering (SISC) segments a univariate series into variable-length motifs using dynamic time warping (DTW) [Sakoe and Chiba, 1978] and DTW barycenter-style centroid updates [Petitjean et al., 2011]. Each segment is summarized by a pattern identity p_m , a duration scale α_m , and a magnitude scale β_m . A pattern-conditioned diffusion model then generates concrete segments from motif conditions using a DDPM-style denoising objective [Ho et al., 2020].

The long-horizon behavior is controlled by the pattern-evolution module, which learns

$$(p_m, \alpha_m, \beta_m) \mapsto (p_{m+1}, \alpha_{m+1}, \beta_{m+1}), \quad (2)$$

with a classification loss for the next pattern and regression losses for the next duration and magnitude scales. This separation is useful, but it also creates a sensitive interface: downstream results depend on how the learned state chain is initialized, reset, and sliced into synthetic training blocks.

3 Replication protocol

Assets and splits. The main datasets are daily Yahoo Finance close-price histories for S&P 500, GOOG, and ZC=F. The S&P 500 setup spans 1980–2020 and yields 10,088 trading days and 2,282 post-SISC segments with $K = 14$. GOOG spans 2004–2020 with 3,776 trading days and 733 post-SISC segments with $K = 11$. ZC=F spans 2000–2020 with 4,764 trading days and 629 post-SISC segments with $K = 11$. We follow a chronological train/test design: motif extraction, fitted generator components, standardizers, and downstream predictors are fit only on their allowed training windows, and final scores are computed on held-out real data.

Downstream benchmark. The main downstream benchmark is TATR: Train on Augmentation, Test on Real. The held-out real slice is divided into an initialization portion and a final real test portion. A one-day-ahead LSTM forecaster is trained on the real initialization portion plus Y synthetic trading years, where one synthetic year has 252 days and Y is swept across a fixed grid. The score is mean absolute percentage error (MAPE) on the final real test portion. We use paper-like LSTM settings: one-day ahead, window size 64, hidden size 32, two layers, MAE training loss, Adam with learning rate 10^{-2} , and 100 epochs for the primary S&P 500 runs.

Synthetic rollout variants. We compare three ways to produce the synthetic augmentation path.

- **Independent fixed blocks:** generate independent 252-day blocks from the same fixed initial state.
- **Continuous chunked:** generate one continuous trajectory with the pattern-evolution chain, then split it into 252-day chunks for downstream training.
- **Continuous with scaler refit:** generate a continuous trajectory and refit the downstream scaler on the augmented training series.

These protocols use the same broad data budget but different Markov-chain connectivity. The distinction matters because the pattern-evolution module is the only component that carries long-horizon state across generated motifs.

79 **Compute.** The primary six-seed S&P 500 downstream batch used stored SISC artifacts and
80 PGM/PEM checkpoints and recorded 10,478 seconds on an Apple M4 laptop with 10 CPU cores and
81 16 GB memory. The lightweight drift diagnostics used the same machine: the 500-seed predictive-
82 alpha script recorded 2.83 seconds, and the 1000-seed selection benchmark recorded 32.69 seconds.
83 These times exclude earlier exploratory runs not used for the reported tables.

84 4 Replication results

85 4.1 TATR depends on the synthetic-series protocol

86 Our current S&P 500 evidence has three layers. First, the stored result matrices do not sup-
87 port a monotone-downward TATR claim for the documented authors independent-block protocol:
88 the stored S&P 500 authors curve finishes 208.4% worse than its baseline, while the single
89 continuous-trajectory curve finishes 71.6% better. Second, a one-seed protocol search found that
90 continuous-rollout variants finish 76.5–85.4% better, whereas independent fixed blocks finish 96.5%
91 worse. Third, Table 1 reports the current primary six-seed long batch.

92 In that long batch, continuous protocols reproduce strong paper-like reductions in downstream
93 MAPE, but independent fixed blocks do not. The conclusion is not driven by a single lucky seed:
94 the final continuous-chunked changes range from -86.8% to -55.1% , while the final independent-
95 block changes range from $+37.1\%$ to $+296.0\%$. The contrast is also visible in Figure 1: continuous
96 trajectories eventually cross below the real-only baseline, while independent blocks worsen as aug-
97 mentation increases.

Table 1: Six-seed S&P 500 TATR summary. Percentages are relative to each protocol’s no-augmentation baseline; lower MAPE is better. Continuous trajectory protocols reproduce large best improvements, whereas the independent-block rollout never improves over baseline and finishes much worse.

Protocol	Seeds	Baseline MAPE	Best change	Final change
Continuous chunked, 1-day	6	0.0589	-84.9%	-65.2%
Continuous + scaler refit, 1-day	6	0.0630	-85.4%	-68.2%
Independent fixed blocks, 1-day	6	0.0646	0.0%	$+193.2\%$
Continuous chunked, 5-day	6	0.0810	-85.2%	-77.5%

98 The mechanism is visible in the synthetic price levels. In the long-batch diagnostics, continuous
99 synthetic trajectories start near 1253 and end near 7085 on average, with mean price level 4138.
100 Independent blocks start from the same level but repeatedly reset, ending near 1249 with mean
101 level 1214. The continuous protocol therefore drifts through the scale of the real test period, while
102 independent blocks remain near the early initial level. This explains why the downstream LSTM
103 can benefit from continuous synthetic augmentation while being harmed by independent blocks.

104 TMTR controls make the replication conclusion more conservative. In the six-seed long batch, the
105 reference TMTR curve reaches a best mean improvement of 26.8% but finishes 359.7% worse; a
106 continuous offset-30 control reaches a best improvement of 50.0% and finishes 17.8% better; a
107 continuous offset-50 control finishes 752.0% worse. Thus the evidence is not “synthetic data always
108 helps.” It is narrower: a particular continuous TATR price-level rollout can create large apparent
109 gains, while other augmentation and evaluation choices do not support the same conclusion.

110 We interpret this as protocol sensitivity, not as a claim of misconduct in the original work. The avail-
111 able generation details do not pin down the downstream result uniquely. A reproducible synthetic-
112 finance benchmark should specify the Markov-chain rollout, initialization, block slicing, scaler fit-
113 ting, and warm-up windows as first-class experimental parameters.

114 The GOOG and ZC=F replication boundary is also important. We regenerated the stated real-asset
115 SISC and architecture artifacts for GOOG and ZC=F with $K = 11$, $l_{\min} = 10$, and $l_{\max} = 21$, but
116 a strict author-faithful downstream TATR rerun is blocked by the released S&P-style split logic: it
117 consumes $252 \times 5 = 1260$ held-out initialization points, whereas the GOOG and ZC=F 80/20 held-
118 out windows contain 744 and 963 points. Stored matrices for these assets also contradict the paper
119 trend: GOOG authors finishes 1003.1% worse and ZC=F authors finishes 75.1% worse. We

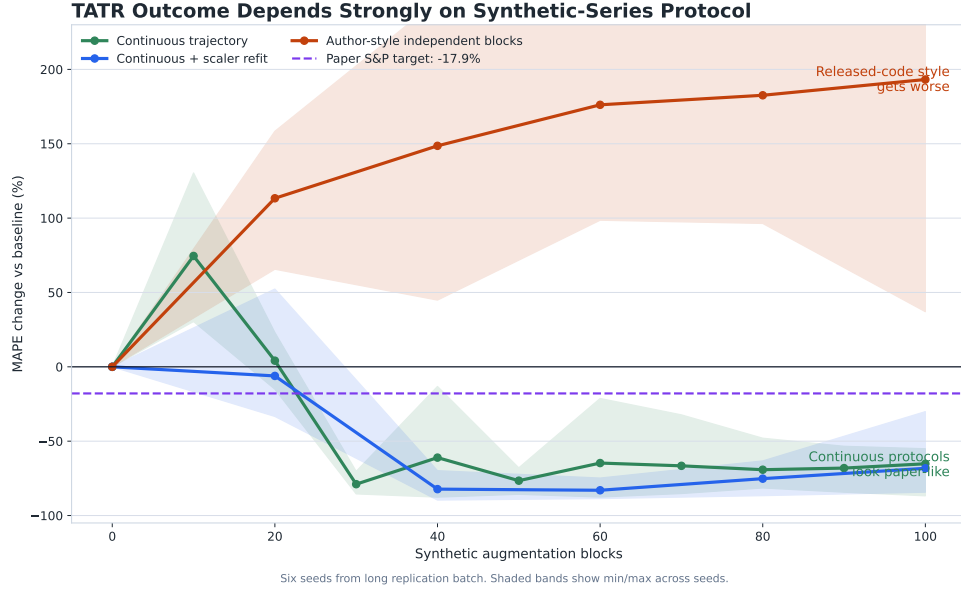


Figure 1: TATR outcome changes sharply with the synthetic-series protocol. Continuous trajectories can look paper-like, while independent blocks worsen with augmentation. The shaded bands show min/max across seeds.

120 therefore treat GOOG and ZC=F as covered for SISC/architecture setup and real-asset diagnostics,
 121 but not as author-faithful downstream replications.

122 4.2 SISC toy-data replication gap

123 We also stress-tested SISC on the Appendix B.3 simulated-data setting. The paper target for the
 124 multi-pattern toy experiment is average per-unit DTW 0.01 and segmentation Jaccard/IoU 0.784.
 125 Across 18 successful runs varying seed, $l_{\max}$, and maximum iterations, our best average per-unit
 126 DTW was 0.0321, and our best author-style interval IoU was 0.6418. Figure 2 summarizes the gap.

Appendix B.3 Replication Gap

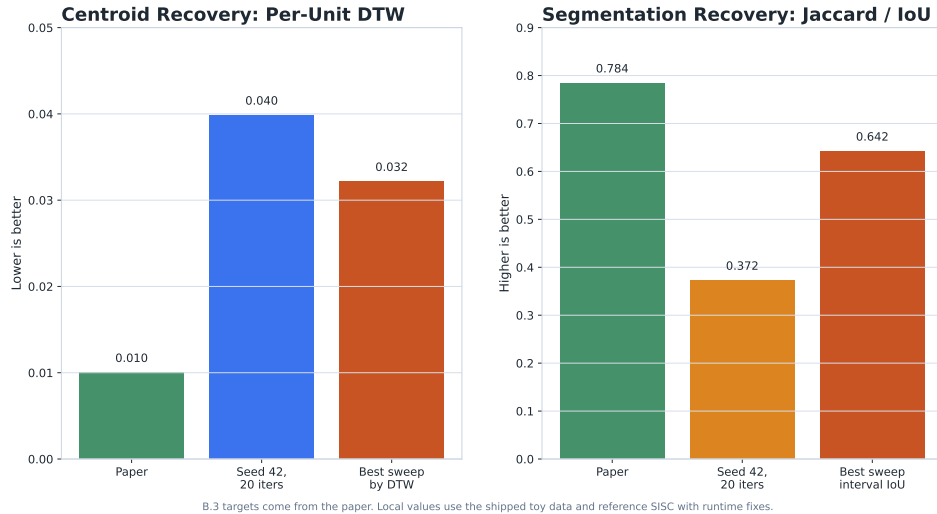


Figure 2: Appendix B.3 SISC replication gap on the toy data. The best replication runs do not reach the reported centroid or segmentation targets.

The gap persisted when increasing SISC iterations from 10 to 20 and when changing $l_{\max}$ from 20 to 21. The best strict boundary Jaccard at tolerance 2 was 0.3608, and the best reference-code-style interval IoU was 0.6418. The best centroid metric remained about $3.2\times$ the reported target. This does not invalidate SISC, but exact B.3 reproduction appears to require more detail about the synthetic generator, seed state, one-pattern construction, and metric definitions. The broader replication lesson is that helper defaults matter: warm-up windows, augmentation scales, plotting conventions, and interval-overlap metrics can all change the apparent result if they are not audited.

5 Predictable-drift extension

5.1 Diagnostic

Stylized facts are necessary but not sufficient for finance. A generator can reproduce heavy tails and volatility clustering while encoding a simple conditional mean, such as $r_{t+1} = \rho r_t + \epsilon_{t+1}$. Marginal tests may not detect this, but a trading signal using r_t can. We therefore treat the drift check as a fixed, predeclared diagnostic rather than a learned discriminator: the chronological split rule, feature map, train-only standardization, null construction, and companion stylized-fact summaries are fixed before evaluating synthetic outcomes.

Motivated by the martingale-difference intuition behind weak-form market efficiency [Fama, 1970], we fix a small information set

$$\omega_t = (1, r_t, r_{t-1}, |r_t|, |r_{t-1}|)^\top \quad (3)$$

and a bounded linear class

$$g_b(\mathcal{F}_t) = b^\top \omega_t, \quad \|b\|_2 \leq 1. \quad (4)$$

The population predictable-drift violation is

$$\delta = \sup_{\|b\|_2 \leq 1} |\mathbb{E}[r_{t+1} b^\top \omega_t]|. \quad (5)$$

Let $m = \mathbb{E}[r_{t+1} \omega_t]$. By Cauchy-Schwarz, Equation 5 has the closed form

$$\delta = \|m\|_2. \quad (6)$$

The empirical estimator is therefore

$$\hat{\delta} = \left\| \frac{1}{T-2} \sum_{t=2}^{T-1} r_{t+1} \omega_t \right\|_2. \quad (7)$$

The statistic decomposes into interpretable components: unconditional drift, first-lag return predictability, second-lag return predictability, and two volatility-conditioned mean terms. In our implementation, returns are standardized using training-split mean and standard deviation only; the constant feature remains equal to one. We compare real data, synthetic data, and a block-shuffled null that preserves the return multiset and some local dependence while disrupting longer chronological alignment.

5.2 Real-asset calibration

Before using $\hat{\delta}$ as a rejection statistic, we calibrate it on real held-out assets. For S&P 500, GOOG, and ZC=F, we start from raw close-price CSVs, use the first 80% of simple returns for standardization, and evaluate the last 20%. The null distribution uses 500 replications of 21-day block shuffling on the held-out returns. Table 2 shows that all three real held-out paths lie inside their asset-specific 5–95% null bands, while retaining heavy tails and volatility clustering. This is a useful sanity check: the diagnostic is not simply rejecting real financial data because it is non-Gaussian.

5.3 Controlled predictive-alpha test

We next test whether larger $\hat{\delta}$ corresponds to exploitable predictability. In each of 500 seeds, we simulate a zero-mean heavy-tailed GARCH-like path, keep the evaluation-period magnitude path $|r_t|$ fixed, and replace only the signs. The iid-sign control has no intentional sign predictability;

Table 2: Real-asset predictable-drift calibration. The null band is the 5–95% range from 500 block-shuffled held-out paths; the final columns report stylized-fact context.

Asset	Test obs.	Real $\hat{\delta}$	Null 5–95%	ACF($ r $)	Ex. kurt.
S&P 500	2018	0.0297	[0.0222, 0.0303]	0.2165	3.2110
GOOG	755	0.0575	[0.0403, 0.0629]	0.0935	6.1142
ZC=F	953	0.0173	[0.0150, 0.0259]	0.1250	2.0247

Table 3: Controlled predictive-alpha test over 500 seeds. Magnitude paths are fixed within seed; only sign dependence changes.

Sample	p	$\hat{\delta}$	OOS corr.	Dir. acc.	Sharpe (sd)
Real evaluation path	–	0.0976	0.0004	0.5025	0.07 (0.77)
IID signs	–	0.0941	0.0036	0.5020	0.09 (0.74)
Markov signs	0.65	0.2088	0.1482	0.5968	2.33 (0.96)
Markov signs	0.75	0.3435	0.2786	0.6952	4.80 (1.05)
Markov signs	0.85	0.5126	0.4112	0.7976	7.72 (1.18)
Markov signs	0.90	0.6164	0.4811	0.8483	9.43 (1.33)

165 Markov-sign samples have sign persistence p . We fit a ridge linear predictor on the first half of each
 166 evaluation path using the same feature map ω_t , then evaluate one-step prediction and a sign strategy
 167 on the second half.

168 Table 3 and Figure 3 show the result. The iid-sign and real controls have near-zero out-of-sample
 169 correlation, near-50% directional accuracy, and near-zero strategy Sharpe. At $p = 0.85$, the hidden-
 170 drift sample has mean $\hat{\delta} = 0.5126$, out-of-sample correlation 0.4112, directional accuracy 79.76%,
 171 and mean annualized Sharpe 7.72 with seed standard deviation 1.18. Across all controlled samples,
 172 pooled $\hat{\delta}$ and out-of-sample Sharpe have correlation 0.6921. Thus the diagnostic is not merely a
 173 reporting statistic; it is aligned with artificial alpha.

174 5.4 Incorporated selection benchmark

175 The FTS-Diffusion use case is direct: report Equation 7 on ordered samples from each rollout pro-
 176 tocol, and do not select a protocol only because it improves TATR if its $\hat{\delta}$ lies outside the real/null
 177 band. We tested this recommendation in a controlled selection benchmark. For each of 1000 seeds,
 178 all candidate augmentations shared the same magnitude path but had different sign dynamics. A
 179 marginal stylized selector minimized a distance over volatility, tail, quantile, and absolute-return
 180 autocorrelation summaries; the drift-incorporated selector minimized

$$d_{\text{styl}} + 0.25 \left[\hat{\delta} / q_{.95}^{\text{real}} - 1 \right]_+^2, \quad (8)$$

181 where $q_{.95}^{\text{real}}$ is the real block-shuffle null cutoff. The selected augmentation trained the same one-step
 182 ridge predictor on a 252-day real initialization plus synthetic returns, and performance was measured
 183 on a final held-out real path that the drift-penalized selector does not see.

184 Table 4 shows that incorporating the diagnostic changes the selected augmentation, not just the
 185 reporting. Relative to stylized-only selection, the drift-penalized selector reduces high-persistence
 186 choices from 39.3% to 0.2%, cuts real-null violations from 76.0% to 34.6%, and lowers held-out
 187 normalized MSE by 0.0365 on average, with a 60.8% paired win rate. If the statistic is incorporated
 188 during training rather than selection, use it as a validation-selected penalty

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FTS}} + \lambda \hat{\delta}_{\text{synthetic}}^2. \quad (9)$$

189 The penalty must be computed on ordered return sequences. Applying it across randomly shuffled
 190 mini-batches would destroy the time ordering and should only be logged as a proxy, not as the final
 191 diagnostic.

Table 4: Drift-incorporated augmentation selection over 1000 seeds. Lower normalized MSE is better. High- p is the fraction of selected Markov-sign candidates with persistence $p \geq 0.75$. The oracle uses held-out test MSE and is only a reference upper bound.

Selector	Null viol.	High- p	Styl. score	Test NMSE	Excess
Oracle test-MSE	41.5%	0.5%	0.2135	0.9979	-0.0319
Drift-penalized	34.6%	0.2%	0.0520	1.0087	-0.0211
Stylized-only	76.0%	39.3%	0.0344	1.0452	+0.0154
Random	84.8%	50.2%	0.3106	1.0579	+0.0281

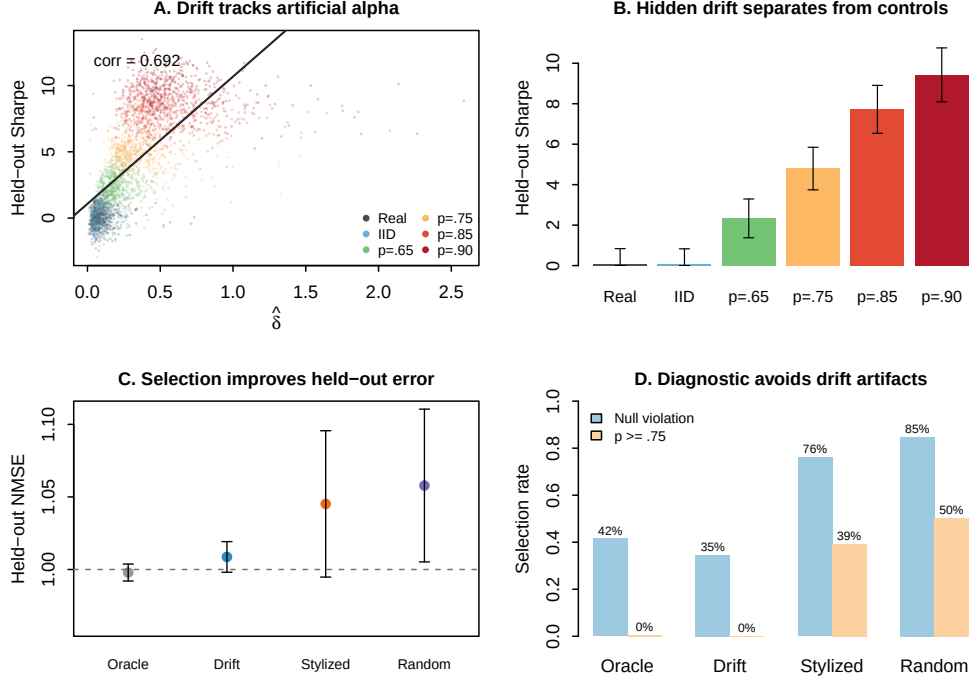


Figure 3: Drift versus no-drift evidence from the larger controlled experiments. The diagnostic tracks held-out artificial alpha, separates hidden-drift sign processes from real and iid controls, and improves augmentation selection by avoiding null violations and high-persistence candidates.

192 6 Limitations and broader impact

193 Our replication is constrained by the information and compute available for this study. Some runs
 194 use reference checkpoints rather than a full fresh retraining of all modules. The drift extension
 195 is calibrated on real assets and validated in controlled predictive-alpha and augmentation-selection
 196 experiments, but the full FTS-Diffusion drift audit and train-time validation selection of λ remain
 197 to be run on ordered generated samples from every rollout protocol. GOOG and ZC=F have shorter
 198 histories than S&P 500, so the downstream split used for S&P 500 is not directly portable without
 199 changing the question being asked.

200 The predictable-drift diagnostic is deliberately narrow. It tests a fixed finite-dimensional feature
 201 map, not every admissible trading strategy, and it is not a proof of no-arbitrage. Its strength is
 202 reproducibility: the statistic is simple, closed-form, and hard to tune after seeing the result.

203 Synthetic financial data can be beneficial for stress testing, augmentation, and reproducibility, but it
 204 can also create misleading evidence if artificial predictability is mistaken for real market structure.
 205 Any release of synthetic paths should clearly label them as generated data, document the training

206 period and generation protocol, and avoid implying that downstream forecasting gains translate into
207 tradable performance.

208 7 Conclusion

209 FTS-Diffusion has the right inductive bias for financial time series: variable-length motifs, pattern-
210 conditioned diffusion, and learned motif evolution. Our replication suggests that the architecture
211 is more robust than the downstream evaluation protocol. Continuous synthetic rollouts can create
212 strong TATR improvements, while independent fixed blocks can fail badly under the same broad
213 data budget. This protocol sensitivity motivates a stricter evaluation standard. Future synthetic-
214 finance benchmarks should report not only stylized facts and forecasting improvements, but also
215 exact rollout protocols and direct predictable-drift diagnostics.

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